WORKED SOLUTION

STEP III 2002, Question 12

Probability and Statistics • Expectation by Conditioning

STEP III 2002, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q12

The key idea

Condition on how the game *starts*. Every game begins with H, with TH, or with TT. In the first two cases nothing useful has been achieved (the run of tails has been broken), so the game simply starts afresh, and its expected remaining length is E again. This "renewal" gives a linear equation for E that we can solve. The same idea, splitting by when the first head appears, handles r tails.

Solution

Throughout, p is the probability of a head, $q = 1 - p$ is the probability of a tail, and tosses are independent. We assume $0 < p < 1$, so that the game ends with probability 1 and every expectation below is finite.

Explaining the equation for E

Look at the first one or two tosses. Exactly one of these happens:

- **H first** (probability p). One toss has been used and there is no run of tails in progress. Because the tosses are independent, the rest of the game is a fresh copy of the whole game, with expected length E. Expected total: $1 + E$.
- **T then H** (probability qp). Two tosses used, and the head has again wiped out the run of tails, so the game restarts. Expected total: $2 + E$.
- **T then T** (probability q^2). The game ends after exactly 2 tosses.

These three cases are exclusive and exhaustive (their probabilities add to $p + qp + q^2 = p + q = 1$). By the law of total expectation,

$$E = p(1 + E) + (1 - p)p(2 + E) + 2(1 - p)^2,$$

which is the given equation.

Finding E

Collect the E terms:

$$E\,(1-p-qp) = p + 2qp + 2q^2.$$

On the left, $1-p-qp = q-qp = q(1-p) = q^2$. On the right, $p+2q(p+q) = p+2q = 1+q$. Since $q \neq 0$,

$$E = \frac{1+q}{q^2} = \frac{2-p}{(1-p)^2}.$$

Expected winnings

Let W be the expected number of heads, which is the expected winnings in pounds. Use the same three cases. After H we have won £1 and the game restarts; after TH we have also won £1 and the game restarts; after TT we have won nothing and the game is over. So

$$W = p(1+W) + qp(1+W) + q^2 \cdot 0.$$

Ask yourself Why is it $1+W$, not W, in the TH case?

The second toss in that case *is* a head, so it earns £1 before the game restarts. Forgetting it is the easiest way to go wrong here.

Collecting terms, $W(1-p-qp) = p(1+q)$, so $Wq^2 = p(1+q)$ and

$$W = \frac{p(1+q)}{q^2} = \frac{p(2-p)}{(1-p)^2} \text{ pounds.}$$

Notice that $W = pE$: on average a proportion p of the tosses are heads, even though the game stops at a moment that depends on the tosses.

The game with r successive tails

Now let E be the expected number of tosses when the game stops at r successive tails. Condition on the first head. Either the first r tosses are all tails, or the first head appears at toss $j+1$ for some j with $0 \leq j \leq r-1$:

- j **tails then a head** (probability $q^j p$). That uses $j+1$ tosses, the run of tails has been broken, and the game restarts. Expected total: $j+1+E$.
- r **tails** (probability q^r). The game ends after exactly r tosses.

So

$$E = \sum_{j=0}^{r-1} q^j p\,(j+1+E) + rq^r.$$

The coefficient of E on the right is $p\sum_{j=0}^{r-1} q^j = p \cdot \dfrac{1-q^r}{1-q} = 1-q^r$, since $1-q = p$. Moving it to the left,

$$q^r E = p\sum_{j=0}^{r-1}(j+1)q^j + rq^r. \qquad (*)$$

? Ask yourself How do we sum $\sum(j+1)q^j$ without a formula sheet?

Write $S=\sum_{j=0}^{r-1}(j+1)q^j = 1+2q+3q^2+\cdots+rq^{r-1}$ and subtract $qS = q+2q^2+\cdots+(r-1)q^{r-1}+rq^r$:

$$(1-q)S = 1+q+q^2+\cdots+q^{r-1}-rq^r = \frac{1-q^r}{p}-rq^r.$$

Hence $pS = \dfrac{1-q^r}{p}-rq^r$, and substituting into $(*)$, the rq^r terms cancel:

$$q^rE = \frac{1-q^r}{p}-rq^r+rq^r = \frac{1-q^r}{p}.$$

Therefore

$$E = \frac{1-q^r}{pq^r},$$

as required. For $r=2$ this is $\dfrac{(1-q)(1+q)}{pq^2} = \dfrac{1+q}{q^2}$, agreeing with the first game.

Checks

- With a fair coin, $p=q=\frac{1}{2}$: the first game lasts $E=\frac{3/2}{1/4}=6$ tosses on average and wins $W=£3$, half of E as it should be.
- $r=1$: the game stops at the first tail, so the length is geometric with mean $1/q$; the formula gives $\dfrac{1-q}{pq}=\dfrac{1}{q}$.
- As $q\to 0$ (almost every toss a head) both E and W blow up, as they should.
- We confirmed these formulae exactly with fractions by solving the chain of states "current run of k tails" for $r=1,\dots,7$ and several values of p. A simulation of 2×10^5 games with $p=0.4$ gave a mean of 4.438 tosses (theory 4.444) and winnings £1.772 (theory £1.778) for $r=2$, and 9.076 tosses (theory 9.074) for $r=3$.

Common slips

- In the winnings equation, writing qpW instead of $qp(1+W)$, which gives the wrong answer p/q^2.
- Conditioning on only the first toss: after a single T the game does *not* restart, because one tail of the run has already been banked. That is why the given equation looks two tosses ahead.

Beyond STEP

The argument "once a head appears, the future looks exactly like the start" is the *renewal* or *first-step* method for Markov chains, and it is how expected hitting times are computed in general. The neat fact $W = pE$ is a case of Wald's identity: the expected sum of a random number N of independent, identically distributed terms is $\mathbb{E}[N]$ times the mean of one term, provided N is a stopping time. You can read about both in Grimmett and Welsh, *Probability: An Introduction*, or in Blitzstein and Hwang, *Introduction to Probability*.

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